



TEST REPORT

Job Fanling North New Development Area, Phase 1: Fanling Bypass Eastern Section (Shung Him Tong to Kau Lung Hang)

Report No. SSRPT9024001710

Customer / Customer address CEDD/ND0
15/F Civil Engineering and Development Building, 101 Princess Margaret Road,
Homantin, Kowloon

Job / Contract No. ND/2019/05

Testing laboratory PWRL(SSK) Tel. 2439 9282

Customer ref. No. C5-ST-208

Date of test request 14/11/2024

Lab. file No. MTDRL-65-2085-1-8-1

Date test commenced 18/11/2024

Date sample received 18/11/2024

Test Request No. SSSTE2490025

Date test completed 19/11/2024

Scope of test :

1. Determination of mass per unit length of steel reinforcing bars in accordance with CS2:2012 (Rev. 6) Cl. 6.1 & 6.2
2. Determination of tensile properties of steel reinforcing bars in accordance with BS EN ISO 6892-1:2009 in conjunction with CS2:2012 (Rev. 6) Cl. 6.1 & 6.4 with modification
3. Rebend test of steel reinforcing bars in accordance with CS2:2012 (Rev. 6) Cl. 6.1 & 6.5
4. Determination of the surface geometry (Relative Rib Area Only) of ribbed steel reinforcing bars in accordance with CS2:2012 (Rev. 6) Cl. 6.1 & 6.7 excluding 6.7.3 with modification (for 40-50mm dia.)*

Information and Test Results:

PART A:

Customer sample No.:	S(1-6), S7, S8	Laboratory sample No.:	SSSTE2490025/S1		
Country of origin & mill:	Alliance Steel (M) Sdn. Bhd.	Cast/Heat No.:	42-6-7200	Type:	Straight
Decoiled, straightening method:	---	Grade:	500B	Nominal bar size (mm):	10
Bar pattern code:	MAL21				Drawing ref. No.: PWL/MAL/3

PART B:

1. Determination of Mass per meter, Tensile properties & Bend performance

Specimen No.			Mass per meter run		Yield strength	Tensile strength	Tensile / yield strength ratio	Total elongation at maximum force	Rebend Test
Laboratory	Customer	RFID Tag No. / Electronic Sample ID	Measured	Deviation from nominal mass per metre					
			(kg/m)	(%)	(MPa)	(MPa)		(%)	
A01	---	SA1135173	0.602	-2.4	631	704	1.12	9.2	---
A02	---	SA1135174	0.602	-2.4	640	720	1.13	10.5	---
A03	---	SA1135175	0.602	-2.4	634	712	1.12	11.7	---
A04	---	SA1135176	0.598	-3.1	645	730	1.13	10.8	---
A05	---	SA1135177	0.599	-2.9	649	737	1.14	9.5	---
A06	---	SA1135178	0.600	-2.8	645	727	1.13	9.5	---
A07	---	SA1135179	---	---	---	---	---	---	PASS
---	---	---	---	---	---	---	---	---	---

2. Surface Geometry

Specimen No.			Relative Rib Area
Laboratory	Customer	RFID Tag No. / Electronic Sample ID	
A08	---	SA1135180	0.049

Notes: 1. *The measured length of the spacing of the transverse ribs for 40mm and 50mm steel reinforcing bars is less than 10 rib gaps.

Remarks :

1. Testing condition: ISO 6892-1:2009 B.
2. Gauge length of the extensometer: 100mm

Checked By

Name CHAN Pak-sum
 Post STO/LAB
 Date 19/11/2024

Certified By

Name CHAN Wai-ming
 Post PTO/LAB
 Date 20/11/2024

This test report is issued without alteration to test data and should not be reproduced except in full.

The results of the tests apply to the sample as received unless otherwise indicated.

The results of the tests relate only to the items tested.

Test related details supplied by customer are printed in italic.

For any enquiry on this Test Report, please contact Public Works Laboratories.

Public Works Regional Laboratory (Sham Shui Kok), No.550 Cheung Tung Road, Siu Ho Wan, Lantau Island, N.T.

PWL STE 1.24(a)Rpt Apr 2024

C Eng D (GEO) 2050 Jul 2023

E-REPORT

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3. Rebend test of steel reinforcing bars in accordance with CS2:2012 (Rev. 6) Cl. 6.1 & 6.5
4. Determination of the surface geometry (Relative Rib Area Only) of ribbed steel reinforcing bars in accordance with CS2:2012 (Rev. 6) Cl. 6.1 & 6.7 excluding 6.7.3 with modification (for 40-50mm dia.)*

Information and Test Results:

PART A:

Customer sample No.:	S(1-6), S7, S8	Laboratory sample No.:	SSSTE2490025/S2		
Country of origin & mill:	Jiangsu Yonggang Group Co., Ltd.	Cast/Heat No.:	F24013836	Type:	Straight
Decoiled, straightening method:	---	Grade:	500B	Nominal bar size (mm):	16
Bar pattern code:	CHA86				Drawing ref. No.: PWL/CHA/10

PART B:

1. Determination of Mass per meter, Tensile properties & Bend performance

Specimen No.			Mass per meter run		Yield strength	Tensile strength	Tensile / yield strength ratio	Total elongation at maximum force	Rebend Test
Laboratory	Customer	RFID Tag No. / Electronic Sample ID	Measured	Deviation from nominal mass per metre					
			(kg/m)	(%)	(MPa)	(MPa)		(%)	
A09	---	SA1135182	1.541	-2.4	564	655	1.16	9.2	---
A10	---	SA1135183	1.539	-2.5	569	659	1.16	9.3	---
A11	---	SA1135184	1.546	-2.1	592	678	1.15	9.1	---
A12	---	SA1135185	1.538	-2.6	576	671	1.16	8.9	---
A13	---	SA1135186	1.546	-2.1	585	673	1.15	8.8	---
A14	---	SA1135187	1.542	-2.3	559	654	1.17	9.1	---
A15	---	SA1135188	---	---	---	---	---	---	PASS
---	---	---	---	---	---	---	---	---	---

2. Surface Geometry

Specimen No.			Relative Rib Area
Laboratory	Customer	RFID Tag No. / Electronic Sample ID	
A16	---	SA1135189	0.087

Notes: 1. *The measured length of the spacing of the transverse ribs for 40mm and 50mm steel reinforcing bars is less than 10 rib gaps.

Remarks :

1. Testing condition: ISO 6892-1:2009 B.
2. Gauge length of the extensometer: 100mm

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